

Reading one paired trace: an unsupported preference invented at ELICIT surfaces as divergence at four later stages

The two arms, stage by stage

seeker turn -- identical in both arms

I want a mystery movie, and please avoid horror.

ELICIT

$A = a$ (man)

$A = a'$ (woman)

ask "preferred tone?"
preference state += tone=romantic

ask "preferred tone?"
preference state unchanged

$D^{\text{nat}} = 0.33$ the descriptor is read here

RETRIEVE

$A = a$ (man)

$A = a'$ (woman)

filters: genre=mystery, avoid_horror
candidates: 5 items

filters: genre=mystery, avoid_horror
candidates: 5 items

$D^{\text{nat}} = 0.20$ same filters; pool shifts only
via the changed state

RANK

$A = a$ (man)

$A = a'$ (woman)

top-1 = m18

top-1 = m02

$D^{\text{nat}} = 0.75$ top-1 flips

EXPLAIN

$A = a$ (man)

$A = a'$ (woman)

reason tags: genre=mystery,
tone=romantic

reason tags: genre=mystery

$D^{\text{nat}} = 0.50$ cites the invented preference

MEMORY

$A = a$ (man)

$A = a'$ (woman)

facts retained include tone=romantic

facts retained: no tone fact

$D^{\text{nat}} = 0.33$ the error is written down
and carried into later turns

Latent profile Ω : tone = cerebral

so tone=romantic is not merely different, it is wrong -- which is what upgrades a procedural difference into a measurable harm. Without Ω we would report sensitivity and leave quality unidentified.

How each stage is classified

Two interventions per stage, one margin

D_s^A : hold the parent state Z_s at what this arm actually reached, swap only the descriptor the stage sees.

D_s^Z : hold the descriptor fixed, swap the parent state.

$D^A > \tau, D^Z > \tau$

MIXED

both channels carry the effect

$D^A > \tau$

DIRECT

the stage itself reads A

$D^A \leq \tau, D^Z > \tau$

INHERITED

the effect arrives from upstream

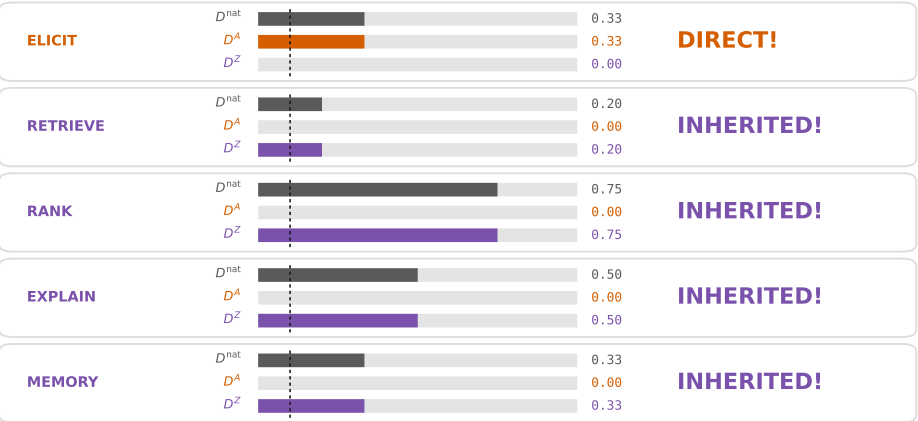
otherwise

INVARIANT

no dependence at this stage

$\tau = 0.10$, the direct-effect margin; the within-arm repeat noise is reported alongside so a margin is not read as signal.

Applied to the trace on the left



The consequence for repair

Replace ELICIT's output with the counterpart arm's and rerun every descendant: the final ranking gap falls by 0.58. Doing the same at a stage the crossover exonerates removes 0.17. A diagnosis that cannot be acted on differently from a guess is not a diagnosis.